

RECENT OPERATING EXPERIENCE AT THE ENRICO FERMI ATOMIC POWER PLANT

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1. Introduction

The Enrico Fermi Atomic Power Plant is presently in the midst of the power escalation programme which has been planned to demonstrate proper operation of the plant at 200 MW(th). As of May 1, 1966, the maximum power level reached has been 67 MW(th). Reactor operation has been satisfactory in all respects. Great difficulty has been experienced in stopping small leaks from the water to the sodium side of the steam generators. Since it is desired to repair leaks before damage to adjacent structures can occur, the steam generators have been shut down frequently when the level, or rate of change of level, of hydrogen concentration in the sodium cover gas has exceeded established limits. Various repair techniques are being tried and some understanding has been gained of the behaviour of the hydrogen in the cover gas as a function of temperature.

It is planned to continue the programme of tests, with achievement of 200 MW(th), the rated power of the present U-10 w/o Mo core, scheduled for July 1966. Several months are to be devoted to tests at 200 MW(th). Following these tests, modifications will be made to the reactor instrumentation and to the steam and feedwater system to allow continued operation as a fuel irradiation facility at power levels ranging from 110 MW(th) to 130 MW(th), depending on the core loading. Such operation will be at decreased reactor sodium inlet temperature and increased reactor sodium flow. The combined effects of lower power, lower inlet temperature and increased flow will result in a maximum fuel temperature of about 800°F (427°C), which is some 300°F (167°C) lower than the maximum fuel temperature existing under 200 MW(th) conditions. This reduced fuel temperature is expected to increase fuel life considerably. Depending on the results of a fuel inspection programme it is possible that as many as 400 days of operation can be achieved before the supply of U-Mo core subassemblies is exhausted.

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During this phase of operation, up to fifteen special fuel test subassemblies, containing a wide variety of fuel and material test specimens, will be irradiated in the core. It is hoped that information obtained from the irradiation of these specimens will be of general importance to the fast reactor community and, specifically, will enable the fabrication of a mixed plutonium oxide-uranium oxide core for the Fermi plant.

The most recent general survey paper on operations at the Enrico Fermi Atomic Power Plant bore the date March 1, 1965.⁽¹⁾ The low power nuclear test programme, which had been virtually completed as of that date, has

TABLE 1. OPERATING DATA SUMMARY

Item	Number completed through May 1, 1966
Reactor start-ups	367
Scheduled scram tests (includes pre-start-up test scrams)	656
Unscheduled scrams	48
Critical reactor lattice configurations	136
Maximum reactor power, MW(th)	67
Subassemblies moved by offset handling mechanism	~ 1500
Offset handling mechanism movements in azimuth	~ 7000
Plug rotations	~ 5000

since been completed. The objectives of this programme were met satisfactorily and complete data were obtained with smaller experimental errors than were anticipated. The results of this programme have been reported in detail in a recent paper⁽²⁾ and it will suffice to say no more here than that the observed physics behaviour of the reactor correlated with the calculated behaviour to a highly satisfactory degree.

On December 17, 1965, regulatory approval from the USAEC was received permitting initiation of the test programme leading to 200 MW(th) operation. This programme was begun on December 29 when the first modest step was taken to 1.3 MW(th).

The present paper will attempt to present highlights of recent operations in such a way that, read together with the references, a complete summary of Fermi operating experience up to May 1, 1966, will be available. Some pertinent operating data are presented in Table 1.

2. Results to Date of High Power Nuclear Test Programme

The Fermi high power nuclear test programme is designed to demonstrate operation of the reactor and associated equipment at 200 MW(th). To achieve this rated power level for the core A fuel loading, a series of eight power levels are scheduled. These power levels were established at 1%, 10%, 50% and 100% of the rated power. Since the Fermi reactor is designed to operate either on two or three coolant loops at a constant flow per loop,

the power escalation programme also incorporates alternate operation under two-loop flow at the above fractional and full power values. Table 2 lists the scheduled power levels and associated flows for this programme.

The primary goals of the high power test programme are to demonstrate the satisfactory operation of all components and systems and to determine the nuclear characteristics of the reactor under conditions of heat generation. The physics measurements include determination of the reactor transfer function, the power coefficient of reactivity and procedures to obtain a satisfactory power calibration by either nuclear or thermal means. In addition,

TABLE 2. POWER ESCALATION PROGRAMME

Power, MW(th)	Flow
0	Full
1.3	Two-thirds
2.0	Full
13.3	Two-thirds
20.0	Full
66.7	Two-thirds
100.0	Full
133.0	Two-thirds
200.0	Full
Full flow	$= 8.85 \times 10^6$ lb/h (4.02×10^6 kg/h)
Two-thirds flow	$= 5.90 \times 10^6$ lb/h (2.68×10^6 kg/h)

a number of tests will be performed to verify the dynamic performance of the reactor and associated systems under conditions of automatic control and the response to power demand and power setback signals. Likewise, the reactor response to loss of a single primary coolant loop and transients resulting from a reactor scram will be studied.

At the present time, tests are being conducted at the 67 MW(th) level with two-loop operation at two-thirds rated core flow, resulting in one-half the full power temperature increase. Sixty-seven MW(th) operation was first achieved on April 1, 1966. Subsequently transfer function measurements have been made under both manual and automatic control modes of operation and a limited number of system transient runs have been completed. A comprehensive summary of the low power physics measurements and the high power tests through 20 MW(th) has been reported in ref. (2). Hence, this section on reactor physics and thermal performance will be restricted primarily to those measurements conducted subsequent to completion of the 20 MW(th) operation.

Transfer Function Measurements

The reactor transfer function in the Fermi reactor is being measured by means of the oscillator rod technique. A rotating oscillator rod with peak-to-peak amplitude of 5.5 inhours is located in a safety rod position midway

between the core axis and the core-radial blanket interface. This rod contains an eccentrically located boron carbide cartridge enriched in boron-10. The rod is rotated at frequencies between 0.005 c/s and 5 c/s and the consequent power perturbation is determined by means of a B-10 ion chamber located in the graphite shield external to the reactor vessel. The ion chamber signal and an electrical signal taken directly from the rotating rod are monitored by a synchronous transfer function analyser which determines the transfer function by cross-correlation methods.

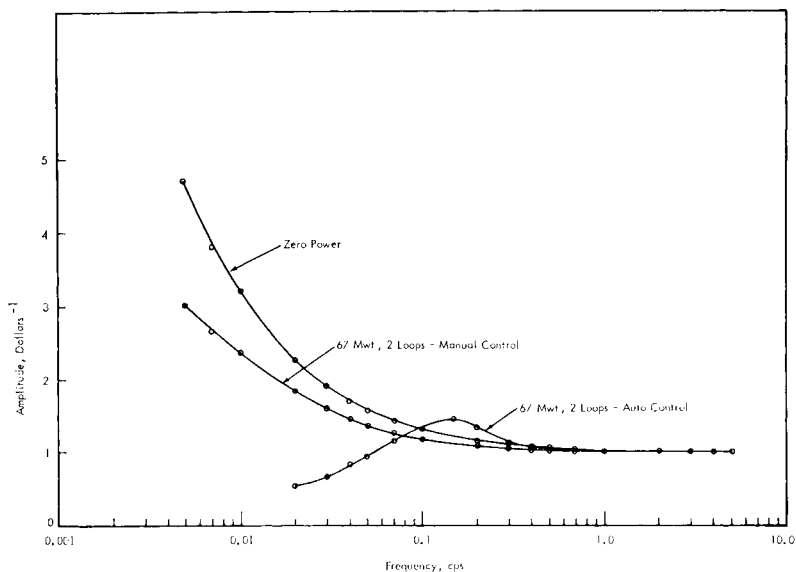


FIG. 1. Amplitude of reactor transfer function.

To date, measurements have been obtained at essentially zero power (constituting all runs through 2.0 MW(th)), and at 20, 30 and 48 MW(th) under three-loop flow. In addition, measurements have been obtained at 13.3 and 67 MW(th) under two-loop flow. As previously noted, the 67 MW(th) condition with two-loop flow corresponds to approximately 50 per cent of rated power level for this flow condition, resulting in a bulk sodium temperature rise of 125°F (69.5°C) with an outlet temperature of approximately 660°F (349°C). Figure 1 represents the measured zero power and 67 MW(th) transfer function amplitudes under both manual and automatic control conditions. Figure 2 is an Argand plot of the power override, corresponding to the product of the power and the power coefficient ($H = PX$). Since the power override, H , does not extend into the first quadrant at high frequencies, it appears that the feedback can be adequately represented by first order time constants. In addition, since no loops are apparent in the low frequency

portion of the power override, it is apparent that transport lag effects are unimportant in the reactivity feedback.

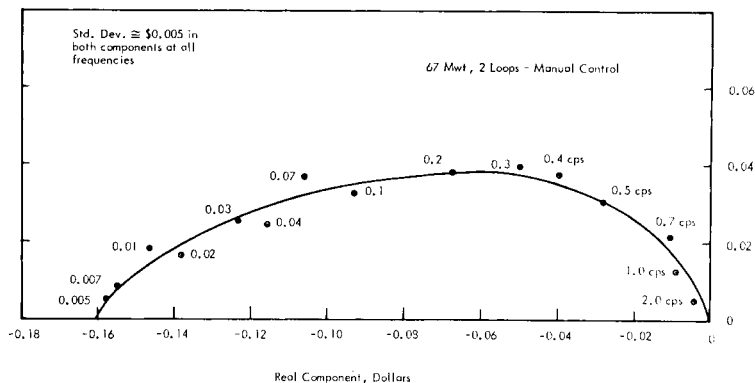


FIG. 2. Argand plot of H .

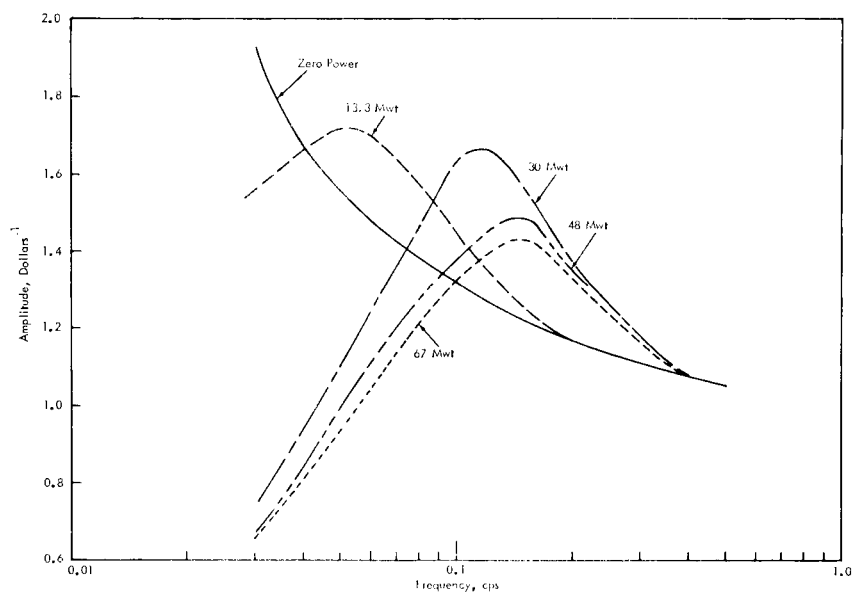


FIG. 3. Measured transfer function amplitude with automatic reactor control.

Preliminary values of the time constants associated with the power override as obtained from measurements at 67 MW(th) are as follows:

15% of H has $\tau \approx 4.0$ sec

50% of H has $\tau \approx 0.5$ sec

35% of H has $\tau = < 0.5$ sec

These time constants are in general agreement with the feedback mechanisms predicted for the Fermi reactor. However, the amplitude is somewhat less than that which had been predicted. This is not totally unexpected, as will

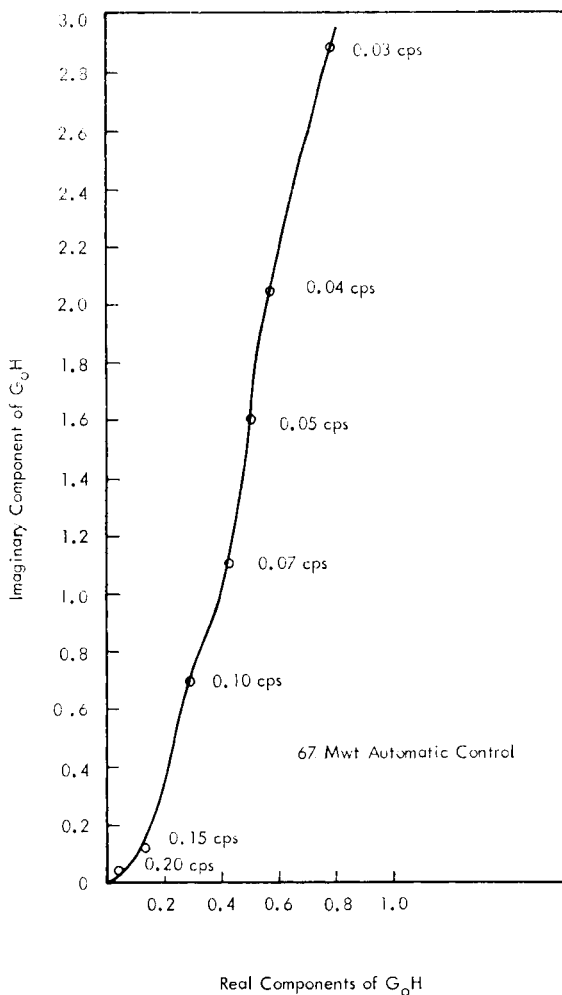


FIG. 4. Plot of G_0H .

be noted later in the discussion of the power coefficient measurements, since the measured isothermal temperature coefficient was also smaller than predicted, i.e. -0.82 ih/ $^{\circ}\text{F}$ (-1.48 ih/ $^{\circ}\text{C}$) versus -0.93 ih/ $^{\circ}\text{F}$ (-1.67 ih/ $^{\circ}\text{C}$).

At each of the power steps achieved to date, the transfer function has been measured first with the reactor under manual control and subsequently with the reactor under automatic control. Under manual control, no effort

is made to use the regulating rod to compensate for the oscillator rod reactivity perturbation. During tests with automatic control, the regulating rod, driven by the time derivative of the neutron signal, attempts to compensate for the oscillator rod reactivity perturbation. The amplitudes measured at 67 MW(th) under both reactor manual and reactor automatic control are shown in Fig. 1. Under manual control, the measured amplitude is less than the zero power amplitude over the entire frequency range, indicating that the reactor is stable. Under automatic control, the amplitude is decreased even more at low frequency due to reactivity compensation by the control system. However, at frequencies near 0.1 c/s, the amplitude exceeds the zero power amplitude, indicating a potentially unstable situation. This characteristic was examined in detail by inserting intermediate steps in the power escalation programme. The resulting amplitude data are shown in Fig. 3. It can be seen that, contrary to the behaviour of a linear feedback mechanism proportional to power, the amplitude peak decreases with increasing power. A plot of G_0H at 67 MW(th) with the reactor in automatic control, shown in Fig. 4, indicates that at these power levels there is a phase margin for stability of approximately 45 degrees. This results from a consideration of the Nyquist Criterion that $G_0H = 1$ for instability, i.e.

$$\frac{G_0}{1 - G_0H} = G_p$$

where G_0 = zero power transfer function, G_p = at-power transfer function.

Although of considerable interest to the reactor physicist at the time, this characteristic was shown to be quite normal for the behaviour of the closed-loop automatic control system utilized for the Enrico Fermi reactor. In fact, it was determined that the control system had indeed been correctly tuned and was operating properly.

Power Coefficient Measurements

The power coefficient of reactivity has been discussed in detail in ref. (2) but will be included here for completeness. It has been measured over the range from zero to 13.3 MW(th) with two-thirds flow and for the range from zero to 20 MW(th) with full flow. Measurements have not been completed for the 67 MW(th) level with two-thirds flow. For each range, critical rod positions are determined under thermal equilibrium conditions at 500 kW, in the mid-power range, and at the upper power level. This procedure was performed during a power descent as well as during a power ascent. The reactivity differences were corrected for changes in coolant inlet temperature and fitted to a straight line. The results are shown in Table 3 and in Fig. 5. The power coefficient in the range from zero to 13.3 MW(th) with two-thirds flow, indicates an 8% difference between power ascent and power descent;

however, this difference is well within experimental error. This result indicated the possibility of a small hysteresis effect in the variation of reactivity as a function of power similar to that which has been seen in EBR-II. However,

TABLE 3. FERMI POWER COEFFICIENT MEASUREMENTS

Power range (flow)	Mode of operation	Power coefficient (ϵ /MW(th))
Zero–13.3 MW(th) (2 loops)	Power ascent	–0.28
	Power descent	–0.26
Zero–20 MW(th) (3 loops)	Power ascent	–0.20
	Power descent	–0.20

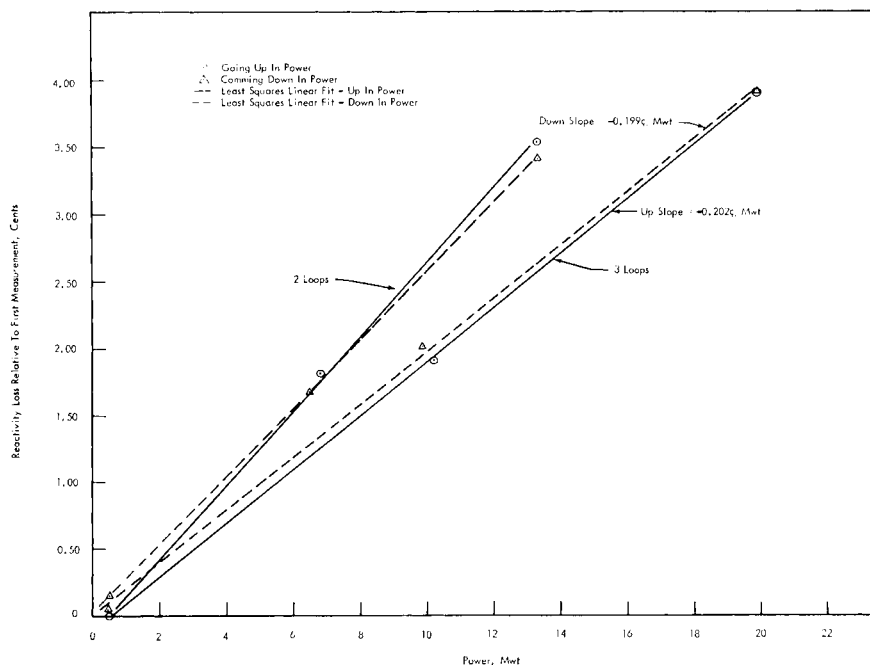


FIG. 5. Power coefficient.

the 20 MW(th) run did not show any indication of such an effect. The uncertainty in the measured power coefficient is estimated to be $\pm 25\%$, principally because an absolute power calibration has not yet been made.

The measured ratio of the two-loop to three-loop power coefficient is 1.40 which is in excellent agreement with the predicted value of 1.41.

The preliminary conclusion obtained from the analysis of the power coefficient data thus far is that the feedback reactivity is approximately 20% smaller than that originally predicted. Since the experimental accuracy is $\pm 25\%$ at these power levels, the difference between measurement and calculation is still quite uncertain. However, a smaller power coefficient would not be completely unexpected since the measured isothermal temperature coefficient was found to be approximately 12% less than the predicted value.

System Characteristics

The determination of the reactor and associated system characteristics is an integral part of the high power test programme for the Enrico Fermi plant. This includes such measurements as the reactor and system response to an automatic approach to power controlled by a programme timer and extending over the range from 5% to 100% of rated power. These tests have been successfully completed to 20 MW(th) under three-loop flow conditions and to 40 MW(th) under two-loop flow conditions. No deviation from predicted performance has been observed.

A limited number of system transients, of types which might occur during normal operation, have been initiated. These include initiation of single circuit shutdown and scram from power. The single circuit shutdown involves the loss of one of the three coolant loops. The Fermi reactor has been designed to accommodate the loss of a single loop without scram. This involves both a thermal and hydraulic transient which must be accommodated by the system. Following the loss of a single loop the change in hydraulic head characteristics results in a flow increase in the two remaining loops. A 25% increase had been originally predicted, whereas an increase of only 15% was observed. A reduction in this value had been expected because the pressure drop across the core has been found to be less than that which was used in the calculational model. In addition, it was generally noted that the thermal time constants involved in single circuit shutdown were considerably longer than had been predicted, of the order of a factor of 2. Likewise, scram transients from 13.3 MW(th) under two-loop operation and 20 MW(th) under three-loop operation indicate a corresponding increase in the thermal time constants as measured from the time of scram to the response of the thermocouples indicating core outlet temperature and to the response of those indicating bulk sodium outlet temperature. Time delays or transport lags have been determined to be approximately as had been predicted, whereas thermal time constants are larger by a factor of 2 in most cases. Detailed analysis is not possible at the power levels investigated to date. More definitive data are expected to result from the tests currently under way at 67 MW(th) and from those scheduled for higher powers.

Thermal Performance

As a result of the difficulties encountered by others in obtaining an accurate nuclear indication of reactor power level, an extensive series of tests was planned to correlate nuclear and thermal power calibration measurements. In the early stages of the low power nuclear test programme for the Fermi reactor, a detailed nuclear power calibration was obtained by means of fission rate distribution measurements, absolute foil activation and the utilization of absolute fission counters. As a result of these measurements, it was determined that an absolute power calibration to an accuracy of better than 10% was available. Subsequent movements and changes in the neutron detectors increased the estimated error to approximately 20%; however, correlation of recent foil activation and heat balance measurements indicates that the nuclear power calibration is still accurate to within 12%.

With these relatively large inaccuracies in mind, a more detailed programme is currently being conducted to obtain accurate thermal heat balance data. Recent measurements at 67 MW(th) have indicated a discrepancy of between 20% and 40% in the electromagnetic flowmeter indications in the secondary sodium circuits. As a result of this unexpected discrepancy in sodium flow information, it has been decided that accurate heat balance measurements will be necessary on the feedwater circuit to enable a subsequent calibration of the secondary and primary sodium flows. Unfortunately, alternative sensors to the electromagnetic flowmeters are not available in the primary or secondary systems. To facilitate the determination of an accurate heat balance, setting loss tests have been conducted with the primary and secondary sodium systems at isothermal temperatures of approximately 517°F (270°C) and 650°F (343°C), corresponding to shutdown and average operating conditions at power, respectively.

An additional and somewhat unexpected result of the thermal performance measurements has been an indication that the thermal effectiveness of the intermediate heat exchangers is lower than expected. This is discussed in the section dealing with the primary sodium system.

Shielding

A preliminary shielding survey was conducted at 67 MW(th) and no shielding deficiencies were observed. This confirms predictions made on the basis of surveys conducted at lower power levels.

3. Experience Related to Steam Generators

In the Enrico Fermi plant three steam generators are used to transfer heat from three intermediate sodium coolant loops to water and steam. The steam generated is supplied to a conventional turbine generator. As is shown in

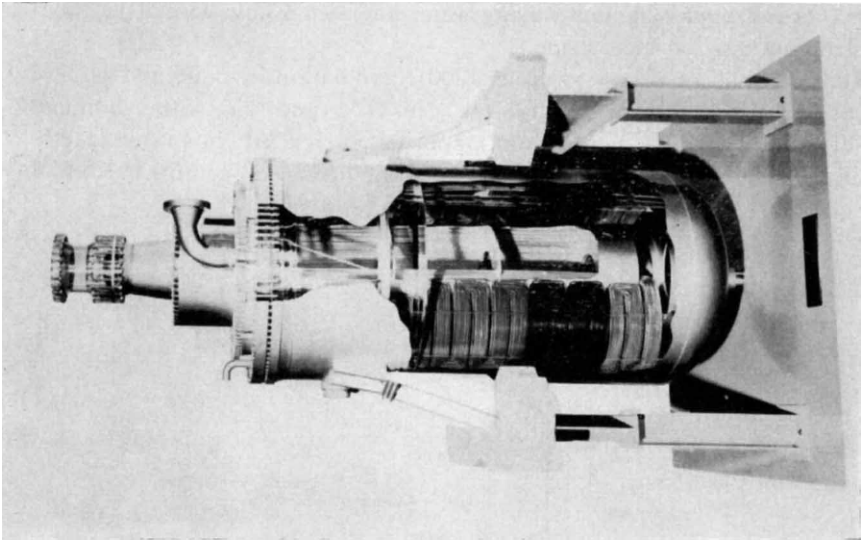
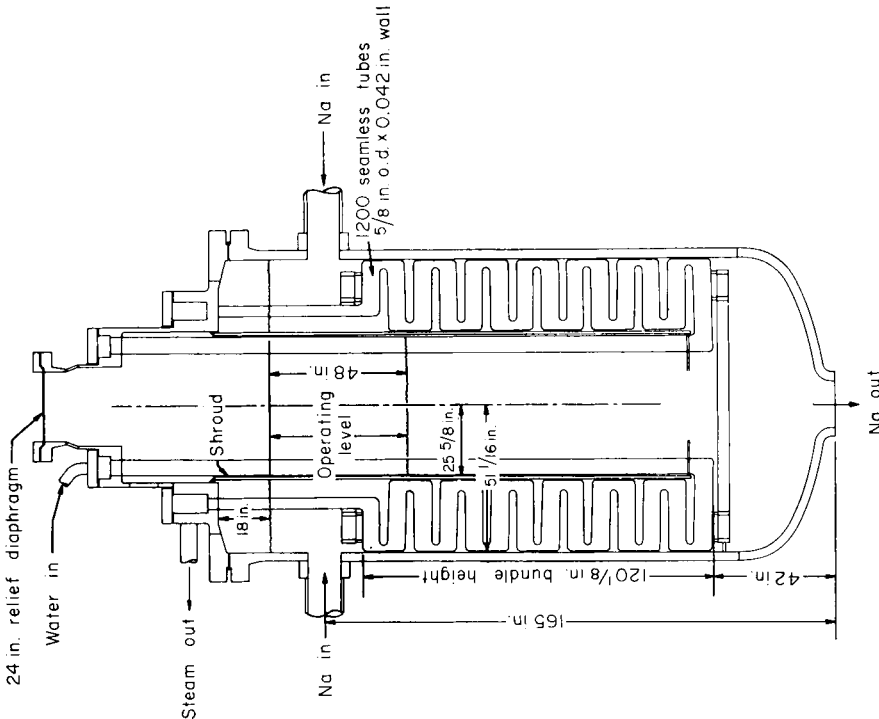


Fig. 6. Model and cross-sectional view of Fermi steam generator.

Fig. 6, each steam generator is a vertical shell, combination cross and counter-flow, once-through type unit having water and steam flow within the tubes and sodium flow on the outside.

The tube bundle is composed of 1200 combination involute and U-bent single wall tubes. Individual tubes are $\frac{5}{8}$ in. (15.9 mm) o.d. with a nominal wall thickness of 0.042 in. (1.07 mm), approximately 80 ft (24.4 m) in length. Each tube is formed to make fourteen horizontal runs between the bottom of

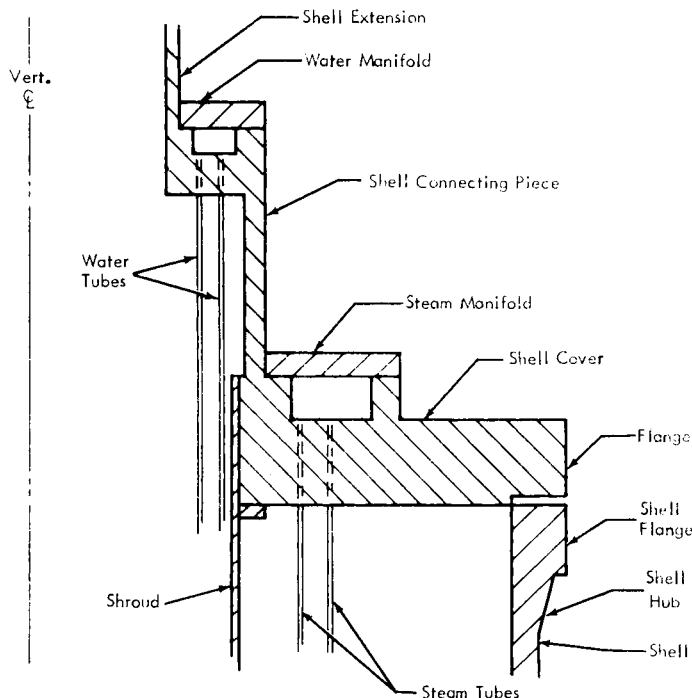


FIG. 7. Detail of water and steam manifolds, Fermi steam generator.

the bundle and the outlet, or steam, manifold. The water side of the tubes is thermally shielded from the sodium inlet temperature by means of a cylindrical shroud which effectively separates the downcomer legs of the steam generator from the steam side.

All tubes, as well as the tube sheet material, is $2\frac{1}{4}\text{Cr-1Mo}$ alloy ferritic steel. The tube-to-tube sheet attachments in the water and the steam manifolds were made by first expanding the tubing into the tube sheets to secure the tubes in position. A fusion seal weld was made between the tube and tube sheet and the balance of the tubing was expanded into the tube sheets.

Sodium enters the steam generator shell through two diametrically opposed inlet nozzles in the upper part of the shell, passes downward across the tubes, and exits at the bottom of the shell. Feedwater enters a ring-shaped manifold

at the top of the shell, passes downward through the vertical downcomer tubes that are enclosed in the thermal shroud, then upward through multiple return bends to the larger diameter ring-shaped steam manifold located below the water manifold. Both manifolds are located in the argon gas space above the operating level of the sodium. Figure 7 shows schematically the manifold arrangement.

The leak detection system employed on these steam generators consists of two continuously operated gas thermal conductivity meters per generator, plus one gas chromatograph which samples the argon cover gas from each of the three loops in 15 min cycles. The presence of hydrogen in the cover gas above the sodium is used as an indication of leakage of feedwater into the sodium.

Considerable experience is required to interpret the meaning of the varying hydrogen levels noted during operation since this level increases with an increase in temperature, particularly as the steam generators enter the superheated steam region. Hydrogen level also increases with an increase in sodium hydroxide and sodium hydride contamination. However, this contamination of the sodium can be greatly reduced by cold trapping the sodium at temperatures of 650–700°F (343–371°C) at intervals as required. It has also been found, as expected, that as the reactor power is raised and feedwater flow is increased, the centre section of the steam generator (within the thermal shield) cools since the temperature of the entering feedwater is 340°F (171°C). This results in an overall drop in hydrogen level because of the larger ratio of gas space and tubing exposed to the gas space in this region compared to the gas volume outside the thermal barrier.

Tests of the hydrogen detection equipment installed on the units have indicated sensitivity to a water leak equivalent to about 0.1 cm³/min. During power operation it has been shown that minor leaks through the tube-to-tube sheet seal welds are quickly noted whether the seal weld slowly deteriorates over a period of days, or rapidly deteriorates, requiring shut down and repair.

Large leaks such as would be represented by tube rupture are accommodated by a system which relieves the hydrogen pressure generated by the sodium–water reaction. This system consists of a rupture diaphragm located in the gas space at the top of the steam generator shell. The total extent of the reaction is limited by an isolating system which shuts off the water to the steam generator and at the same time dumps the water and steam from the unit in approximately 45 sec.

Much of the early history of the Fermi steam generators has been covered in previous papers.^(3, 4) This paper will only briefly review the significant events of interest before March 1965 and will continue on from that date in more detail.

In 1961, prior to being placed in service, pressure tests revealed that many

of the tubes in the No. 2 unit were leaking. It was found that the leakage had been caused by stress corrosion cracking at the tube bends, probably attributable to residual cleaning solution containing sodium hydroxide and sodium nitrate. No. 2 unit was completely retubed in 1962, using the original tube design and material. Subsequently, all three generator tube bundles were partially stress relieved; no further stress corrosion has been discovered.

In November 1962 the No. 1 steam generator was placed in service and operated isothermally at 470°F (243°C). Two weeks later vibration of the tubes directly in front of the sodium inlet nozzles caused a failure of several tubes and a sodium-water reaction occurred. The relief and isolation system functioned properly to limit the reaction. At that time only the gas chromatograph had been provided to warn of a water leak. The chromatograph indication did serve to alert the operator to watch cover gas pressure and the unit was isolated manually. The vibration problem was finally corrected by installation of a baffle in front of the sodium inlet nozzles and by lacing the tubes at two elevations on the outer vertical section of the tube bundle. There have been no further tube failures since these modifications were made on all units.

Although the steam generators were not required for the low power nuclear test programme which began with initial criticality in August 1963, work continued on these units throughout 1964 to make the tube bundles as leak-tight as possible. By February 1965, 270 tube-to-tube sheet seal welds had been repaired on the No. 2 unit and 160 on the No. 1 unit. Most of these repairs were made following dye penetrant testing of the tube sheet welds. All pits and flaws in the seal welds, as disclosed by the dye tests, were ground out and rewelded using 2½Cr-1Mo filler rod and tungsten inert gas welding techniques.

Actual seal weld leaks were located by one or both of two methods. One method consisted of injecting several hundred cubic feet of helium into the gas space of the steam generator shell and searching for helium leaking past the seal welds at the water and steam manifolds. A mass spectrometer was used for this purpose with limited success.

A second method consisted of pressurizing the intermediate sodium loop to about 4.4 atm with argon gas while the tube bundle was filled with condensate to a level just above the seal welds. With the manifold cover plates removed, a visual inspection of each seal weld was made to detect any gas bubbles developing at the welds. Since these leaks are extremely small, it is a tedious and not completely successful method of leak detection.

The final leak rate is determined by pressurizing the tube bundle to 42 atm with nitrogen after repairs have been made and the loop filled with sodium. The gas chromatograph is then used to determine the leak rate by observing any increase of nitrogen level in the argon cover gas. A good repair job will generally limit the nitrogen leakage to 20–30 cm³ per day.

All three steam generators were put in service in March 1965, but were operated under isothermal conditions with only about 1000 lb/h (450 kg) feedwater flow through the units. Sodium flow in the intermediate loops varied between 2 and 3 million lb/h (0.9 and 1.36 million kg) per loop and sodium temperature was maintained at approximately 450°F (230°C). The steam generators were boiled dry on four occasions during the year and each time nitrogen pressure tests at 42 atm were made to determine if any significant leakage had developed in the units. No change was found, indicating that under isothermal conditions, at least, the steam generators were sound. A typical leak rate test in April 1965 gave the following results:

No. 1 steam generator—22 cm³ of nitrogen per day.

No. 2 steam generator—not detectable.

No. 3 steam generator—90 cm³ of nitrogen per day.

Upon receipt of the 200 MW(th) licence, the temperature of the sodium systems was increased to 517°F (270°C), which is the normal reactor start-up temperature, and the first programmed increase in power level to 1.3 MW(th) was conducted on December 29. Subsequent steps in the approach to power programme have included testing at 2, 13.3, 20 and 67 MW(th). At 67 MW(th) on two sodium coolant loops, the temperature of the sodium entering the steam generators is approximately 630°F (332°C), the exit temperature is 517°F (270°C), and the feedwater enters at 340°F (171°C).

As the reactor power level has been raised, accompanied by increased feedwater flow and greater temperature differentials across the steam generators, the problem of maintaining leak-tight steam generators has also increased. No. 3 steam generator has been under repair on two occasions since the first of the year, once after 20 MW(th) and again after 67 MW(th) operation. On each occasion pressure testing indicated a leak rate of about 350 cm³ of nitrogen per day. No. 1 steam generator is presently under repair having been shut down in April after 67 MW(th) operation when its leak rate increased to 3000 cm³ per day. No. 2 steam generator was taken out of service on May 1 and work has begun to eliminate leaks in this unit.

Using dye penetrant, bubble testing, and mass spectrometer techniques, all the leaks which have been found are in the tube-to-tube sheet seal welds. Some cracks in the seal welds of the inner row of tubes in the water manifold have extended into the surface of the ligaments of parent material between tubes. Nine of these shallow cracks, up to $\frac{1}{4}$ in. (6.3 mm) in depth, have been repaired on the No. 3 unit by grinding out the crack and welding back to original dimensions with 2 $\frac{1}{4}$ Cr-1Mo filler rod. The width of the ligament between tubes is about 0.155 in. (3.94 mm) in this inner row and increases in width in successive rows in the radially outward direction. A number of these ligament cracks have also been found in the No. 1 unit and repaired by the same method. None has been seen in the No. 2 unit.

Studies made by the supplier of the steam generators, and also by an impartial consultant, have indicated that the stresses in the region of the inner row tubes are not great enough to cause cracking in normal operation unless there were notches or defects in the tube sheet forging. However, the stress effects are again being recalculated for several operating conditions such as filling of the units with water, operation with very low feedwater flow, and isolation of the units.

The nature of the leaks found at the seal welds is generally found to be that of pinholes probably caused originally by oil or dirt deposits in the tube-to-tube sheet joints. The latest method employed in sealing these pinholes and other defects is to deposit, by tungsten inert gas welding, an overlay of Nicobraz 180,* a nickel brazing material, over the original weld. Metallographic examinations were conducted on laboratory samples after brazing over both sound and porous welds. After thermally shocking the specimens from 800°F (427°C) to 70°F (21°C), the general conclusions were as follows:

1. Some melting of the first weld occurs during brazing, and mixing of the braze metal and steel occurs superficially.
2. A metallurgical bond is produced between the weld and the braze material.
3. Cracks in the seal welds stop at the braze interface.
4. Three thermal shocks from 800°F to 70°F do not cause the braze to lift from the weld.
5. The hardness value of the braze metal, as deposited, is lower than that of the underlying weld metal.

Some 300 tube ends have been brazed on the No. 3 steam generator to test this method of sealing poor welds. Several weeks of operation have disclosed no new leaks in this unit, and the same procedure is now being used to seal approximately 500 welds in the No. 1 unit.

The present maintenance schedule indicates that the No. 1 steam generator will be ready for service the week of May 16. At that time reactor operation will continue at the 67 MW(th) level with Nos. 1 and 3 units in service. The No. 2 steam generator will probably be in repair for several weeks to permit reworking of defective seal welds.

It seems likely that all three generators will require continued attention in the future. Eventually it may be necessary to reweld all of the seal welds.

4. Experience with Other Plant Components and Systems

Reactor Components

Fuel

Fuel performance from the standpoint of radiation resistance has not been evaluated as yet since burnup levels reached to date are low. What has been

* The composition of Nicobraz 180 is 87.4% nickel, 5% chromium, 3.5% iron, 3% silicon, 1% boron and 0.1% molybdenum.

monitored, however, is reactivity of a reference core to determine whether any changes are taking place. It was reported in reference (1) that the reactivity of the reference core had increased 31 inhours over a period of about 400 days, with each significant increase in reactivity occurring after several days of isothermal operation at primary system temperatures of the order of 600–700°F (315–370°C). This phenomenon has continued over the past year so that now the total gain is about 50 inhours.

Most of this gain can be attributed to a phase transformation in the U–Mo fuel with resulting shrinkage. However, the total reactivity gain is too large to be accounted for by this mechanism alone, even if 100% transformation is assumed. To determine whether or not hydriding of the zirconium fuel cladding is responsible for part of the increase, one of the sodium worth subassemblies was removed from the reactor for examination. From this examination, and examination of unexposed depleted pins and depleted uranium pins previously removed from the reactor, it has been concluded that about one-half of the reactivity gain is attributable to phase change and one-half to hydrogen pick-up in the order of 100 ppm in the zirconium clad. Reactivity monitoring will continue in order to observe any significant changes from expected values. Data will not be as precise as in the past because of fuel burnup, but changes will still be observable before significant hydriding of fuel cladding takes place.

Control rods and drives

Control rods and drives have continued to function well. Several minor instances of failed switches and slipping clutches have occurred but repairs have been readily made. Periodic checks of scram time and of forces required to raise and lower the safety and operating control rods have shown consistent performance in operation.

One design change has been made in the safety rod extensions to make it easier to monitor for bellows leaks. Each extension cavity had been sealed and pressurized with argon with a gauge mounted where it could be checked occasionally. The fixed volume resulted in rather large pressure changes with temperature and, with the machinery dome in place, it was difficult to observe the pressure gauges. Therefore, a plastic tube was connected from each extension to a header which was in turn connected to a pressure control system. This then maintains pressure at a constant value, provides a continuous feed in case of failure to prevent sodium from getting into the extension, and provides a continuous failure monitor by use of an excess flow alarm. The major problem in design was providing space for the plastic tubes which would allow them to move up and down with the extension without interference. The resulting operation is simple and effective. In addition to continuous monitoring, a monthly pressure hold check is carried out to determine if very small leaks have developed.

Neutron source

A two-piece antimony-beryllium neutron source is used in the Fermi reactor. Several antimony slugs about 5 in. (12.7 cm) long are clad in a tantalum jacket designed for high temperature operation in liquid sodium. Cladding is done before irradiation which, incidentally, results in higher activity in the tantalum clad than in the antimony. Because of the high clad activity, there has been a troublesome problem in handling the antimony source section when it is removed from the shipping cask under water prior to insertion in the reactor. The water pool and the cleaning chamber, where the source is preheated and dried before placing it in sodium, have been contaminated with micron sized particulates which seem to come from the tantalum surface. Because of this difficulty, a stainless steel clad source is planned for use in the lower temperature 110 MW programme starting in 1967. Before another high temperature core is obtained, consideration will be given to using antimony tetroxide in stainless steel or to a double clad with an inner layer of tantalum and an outer layer of stainless steel. Another alternative is to can the slugs of antimony in tantalum after irradiation rather than before. At present, however, there is not the necessary high degree of confidence that tantalum can be welded satisfactorily by remote control in a hot cell.

Fuel handling

Reactor fuel handling equipment at the Fermi reactor has been operated extensively and continues to operate well. Inside the reactor, the transfer rotor, rotating plug, sweep arm, and offset handling mechanism have had but minor problems. On several occasions, a subassembly has hung up when it was being inserted in the lattice. A second or third attempt at insertion has often succeeded. If not, shutting down pumps for a short period of time to relieve flow forces on the subassembly, or indexing the camming surfaces by insertion in a storage position, has been adequate to correct the difficulty. Maintenance on the equipment has been limited to occasional repair of a motor, relay, or gear box. Fuel handling problems external to the reactor are discussed below.

Reactor Control System*Detectors*

During the low power nuclear test programme, the neutron detectors used were designed for low temperature operation. They were cooled by a stream of air, used low temperature cable, and did not require shield plugs at the top of the guide tubes. For high power operation, permanent detectors with integral cables (connectors only on source range detectors) were installed

with shield plugs. Placing of the detectors had been based on low power flux data.

Operation of the permanent detectors to date has been satisfactory. Intermediate range B-10 detectors are being operated with a low compensating voltage until sufficient gamma flux is obtained to make an accurate adjustment. To be sure that cable leakage currents at high temperature do not interfere with signal currents, intermediate range and source range detectors remain cooled to some degree. Additional testing is planned to determine if cooling can be shut off.

In the past year, two additional source range detectors have been installed to increase source range sensitivity and thus extend source life. This was felt necessary because of the schedule of reactor operation which did not call for extended enough runs at power to significantly reactivate the source. Source range sensitivity was increased by installing two additional fission counters mounted in a spare guide tube and surrounded by cans of ZrH_2 moderator. They can be switched into service when needed because of low source strength, one in parallel with each of the original detectors, and are interlocked so they must be switched out before intermediate range channels are on scale. This is to assure adequate overlap between source and intermediate range channels. The investment in the additional detectors, racks, and ZrH_2 , etc., has already been repaid by reducing source requirements by at least one.

Safety system

The safety system continues to function well. Essentially no difficulty has been encountered with component failures or with calibration changes. The incidence of unscheduled scrams has been considerably reduced from what it was during initial plant operation but is still too high. Part of this is due to the test programme which sometimes involves unusual conditions. Another part has been due to equipment not yet being fully adjusted for power operation. One example of this is the readjustment made in the fast rundown mode of safety rod action. Fast rundown is initiated by a relay which turned out to be much faster than the magnets themselves so that an extremely short noise pulse would initiate a fast rundown but not a scram. The response of this relay has been reduced to a level compatible with other safety actions. This together with other similar changes should reduce the incidence of unscheduled scrams to a low value during normal plant operation.

Control system

Tests of the automatic control system and of the associated items which together make up the automatic approach to power system are described elsewhere in this report. Operation to date has been satisfactory except for two problems with voltage stabilization in timer comparison circuitry.

Asymmetric loading of the three-phase essential bus which feeds the control system, and a voltage regulator with a non-sinusoidal output voltage were the problems. The cause of the unsymmetric load and the voltage regulator were removed to correct the faults.

The reactor is extremely stable with and without the automatic control system, and can be very smoothly brought to power either manually or automatically. A relatively complicated automatic system can probably be justified for a reactor which is started up and shut down often, but is probably unnecessary if start-ups are less frequent such as in a reactor with a high excess reactivity, high burnup core. The automatic system may be very useful at Fermi, but it probably could be eliminated without causing much distress.

Primary Sodium System

Experience in operating the primary sodium system has been good. Some specific comments in this area follow.

Primary sodium pumps

No major difficulties have been experienced during the operation of the three primary sodium pumps. No problems have been observed with the modified check valves which were installed about 2 years ago. The shaft face seals located in the cover gas area have required rather frequent replacement but this does not appear to be due to the fact that the pumps are pumping sodium. Development of improved seals is continuing. Motor speed control on these three pumps is accomplished by varying the external electrical resistance of the rotor circuit. Rheostats containing a water solution of sodium carbonate are used for this service and have been found to operate satisfactorily although with relatively high maintenance requirements. A dry, mechanically simpler system would be investigated if replacement was considered.

Intermediate heat exchanger

Since extensive operation at significant power levels has not yet been achieved, observations on the intermediate heat exchangers have been limited. No obvious problems have developed.

During operation at 67 MW(th) it was possible to obtain some preliminary data on the performance of two of these heat exchangers. These data suggest that the effectiveness is considerably lower than expected. Under the operating conditions the primary side inlet and outlet temperatures were 655°F (346°C) and 545°F (285°C) respectively with corresponding secondary side inlet and outlet temperatures of 520°F (271°C) and 630°F (332°C). The effectiveness factor, calculated as the ratio of the primary side temperature difference to

the theoretical maximum temperature difference measured, i.e. $\frac{655-545}{655-520}$, was found to be 0.81. This should be compared to a predicted effectiveness of 0.884. A calculation of the product of the heat transfer area and the average heat transfer coefficient has been made indicating that this product is 4.2×10^6 Btu/h/°F (4.2×10^6 C.h.u./h/°C) as compared to the predicted value of about 7×10^6 Btu/h/°F (7×10^6 C.h.u./h/°C).

It should be emphasized that the above statements are based on preliminary data. A programme of careful measurements to check these data will be carried out at higher power levels.

Primary sodium chemistry

In the primary system no particular sodium chemistry problems have arisen. There has been no difficulty in maintaining the plugging temperature at 240°F (116°C) or below with only a few hours of cold trap operation per month. Samples of primary sodium are taken monthly for chemical analysis. Characteristic values found upon analysis are total oxygen 10 ppm, total carbon 50 ppm, carbon in sodium carbonate 1 ppm and total hydrogen 1 ppm.

A total of twenty-four stainless steel specimens have been exposed to primary system sodium at 1200°F (650°C) for periods of approximately 1000 hours. These specimens have then been analysed for surface carbon content and all the results so far have indicated the degree of carburization to be less than the conservative value of 0.5% which has been established as a limit.

Primary cover gas

No major operating problems have been encountered in the primary argon cover gas system. Sufficient argon activation has taken place so that any leaks would have appeared but few leaks have been found. The only significant leak occurred at one of the primary pump face seals and was in itself not serious from the radioactive contamination standpoint. A faulty O-ring seal has since been replaced.

The semi-graphite used in the reactor top plug communicates with the primary cover gas. During tests made before nuclear operation it was found that this material degassed at elevated temperature, emitting principally hydrogen and methane. A continuous analysis of the cover gas is made to assure that the concentrations of these gases as well as carbon monoxide and carbon dioxide do not become unduly high. Measurements made during and after operation at 67 MW(th) with a reactor sodium outlet temperature of 650°F (343°C) indicate that no significant increase in plug degassing occurred. This will be watched closely as power and temperature levels increase in order to allow purging before cover gas contaminant concentrations become high enough to result in contamination of the sodium.

The nitrogen concentration in the primary cover gas continues to remain higher than can be readily explained. This value varies between 1000 and 4000 ppm and may be because of a leak into the primary cover gas system from the nitrogen atmosphere in the main reactor shield. Interest in the nitrogen concentration was renewed when it was found that the EBR-II also showed higher nitrogen than expected.

A charged wire fission product detector has been installed and commissioned in the primary cover gas system but no significant data have yet been obtained.

Instrumentation

Level, flow, pressure and temperature instrumentation installed in the primary system has operated fairly well. Difficulty has been experienced in calibrating temperature detectors in the three sodium loops and a good deal of inductive reasoning must be used in interpreting the readings. A recent isothermal elevation of the primary system to 650°F (343°C), together with a careful physical examination of the detector circuitry, was carried out in an attempt to alleviate this condition.

One of the primary electromagnetic flow meters has developed an erratic output which is unexplained to date. It is planned to enter the reactor equipment compartment to examine the magnet installation since it is possible that the magnet may be moving with respect to the pipe or the leads may be improperly attached, thus giving an erratic signal.

Fuel Handling External to Reactor Vessel

The preparation of new fuel and the cleaning and storage of irradiated fuel is accomplished by the fuel cleaning system in a separate building which is close to the reactor building. All core and blanket components, such as fuel and blanket subassemblies, the gamma emitter section of the neutron source, control rods, and safety rods, are handled through the fuel cleaning system and transferred between that system and the reactor in the portable cask car. The entire initial core loading of 105 subassemblies was handled by this combination of equipment. Much experience was gained during the loading programme and significant modifications to the various systems and components have since been made.

Cask car

The cask is a lead and depleted uranium shielded self-contained portable cask. It is designed to handle up to eleven fuel subassemblies at a time. The internal fuel storage rotor, the recirculating argon heating and cooling system

and the hoisting equipment are all an integral part of the car. Operating experiences revealed many design deficiencies. Problems of major importance included: distortion of the cask due to differential growth of the shielding materials at operating temperatures, sodium contamination of cask internals and extreme difficulty in performing maintenance due to the complexity of the car assembly.

A new system to replace the cask car has been designed and is currently being built. The transfer between the reactor and the fuel cleaning system will be made in a portable cask capable of handling one subassembly at a time. The reduced capacity of the cask permits elimination of the recirculating argon system and resultant sodium contamination of car internals. Components are of simple design and perform singular functions. The new design will provide greater flexibility and reliability of operation in addition to significantly improved ease of maintenance.

Fuel cleaning system

The fuel cleaning system consists of three major components: the cleaning machine, a sodium filled transfer tank, and the process system. The cleaning machine very simply described is a tube mounted on a hinge that is capable of rotation to various stations within a shielded cell. The machine is equipped with a gripping device that fits closely to the inside wall of the tube. New fuel is heated inside this tube prior to insertion into sodium. Repeated overheating of the tube in combination with inadequate provisions for expansion led to deformation and loss of clearance between the tube wall and the gripper. These problems have been eliminated through modifications in the design of the tube which provides for expansion and improved heating control.

The cleaning machine removes sodium from components by a steaming process. The effluents from the process are discharged to the process system through connecting piping. At the conclusion of the steaming process the machine and the piping are wet and must be dried prior to connexion to the sodium filled transfer tank. Considerable difficulty was encountered in drying out the system and measuring for dryness. The system was completely repiped to provide better drainage, heating, and purging to accomplish drying. Remote reading dew point indicating instruments have been installed to measure system dryness. The modifications have been completed and initial testing indicates successful resolution of the problem although some difficulties in assuring adequate gas flow through the subassembly have been encountered.

During the loading programme the oxide level of the sodium in the transfer tank continued to increase indicating insufficient capacity of the sodium clean-up system. The oxide build-up was accompanied by a heavy film of oxide over the sodium surface. A new cold trap system has been installed which has increased capacity, and provided for skimming the sodium surface

and for operating at elevated temperature. The new system is currently being tested and initial results indicate satisfactory operation.

Pools

Two stainless steel lined water filled pools for the storage and handling of irradiated assemblies are located adjacent to the fuel cleaning machine. Although no irradiated fuel assemblies have been handled in the pools, some experience has been gained in handling the gamma emitter section of six neutron sources. Overall operations have been quite satisfactory. A problem of reduced visibility through the water due to a build-up of turbidity over a long period of time was eliminated by the installation of a high efficiency filtering system.

Containment

Testing of various parts of the containment system continues on a periodic basis. Valves, bellows seals on piping, air locks, and the equipment door are all tested at full design pressure on a quarterly basis, with the test period extended to 6, 9, and 12 months provided two successive tests are successful without intervening maintenance. In order to test the air locks it was necessary to procure strongbacks to reinforce the inner door because test pressure between doors pressurizes the inner door in the wrong direction. Many of the valves, the air locks, and other components have shown no significant increase in leak rate and have not required maintenance for two successive tests so that the next test has been scheduled on a 6 month basis.

Electrical penetrations are tested monthly at full design pressure of 3.18 atm. All electrical penetrations have now been modified to include a plastic sealant. Since this modification was completed there has been much less difficulty maintaining a tight seal, and the last three monthly tests have been so good that no maintenance has been required.

Overall manpower requirements for building and other containment component testing are quite high. There seems to be no alternative to extensive testing, however, if valves and gaskets are used for securing tight containment and if continued containment integrity is to be assured. The work load, however, could be much reduced if sufficient attention was given to containment during design to minimize penetrations and to provide for ease of testing. Minimizing penetrations would possibly require a larger containment building so that many components of cover gas, waste gas, sodium cleanup and other systems could be inside the building. Thus no penetrations, or perhaps just one penetration, would be required for each such system. Assuring containment integrity of each of many penetrations is a big job, which can be done, but the job can be minimized by careful design.

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